



聖凰科技股份有限公司

SANTA PHOENIX TECHNOLOGY CO., LTD.

美光_EQ LP (PCB Modbus MFC Type.) 教育訓練手冊

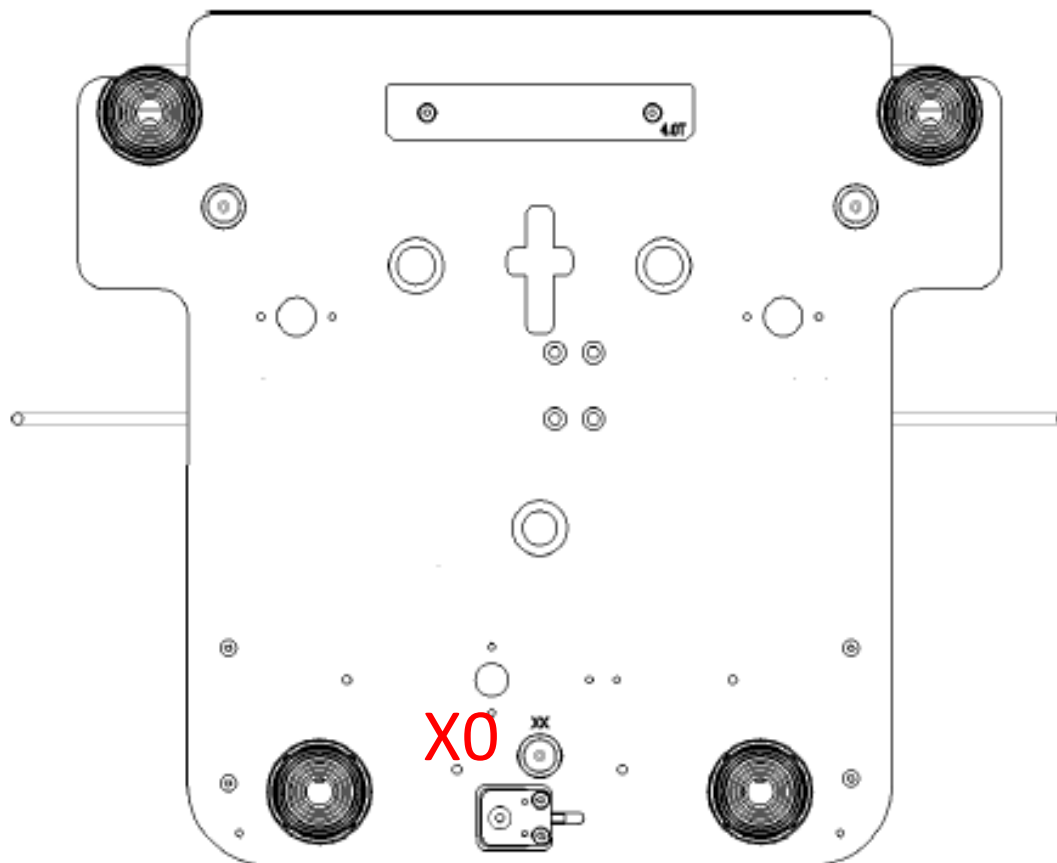
編輯：Hugo

日期：2026.03.03



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美光 EQ_LP LP盤架構





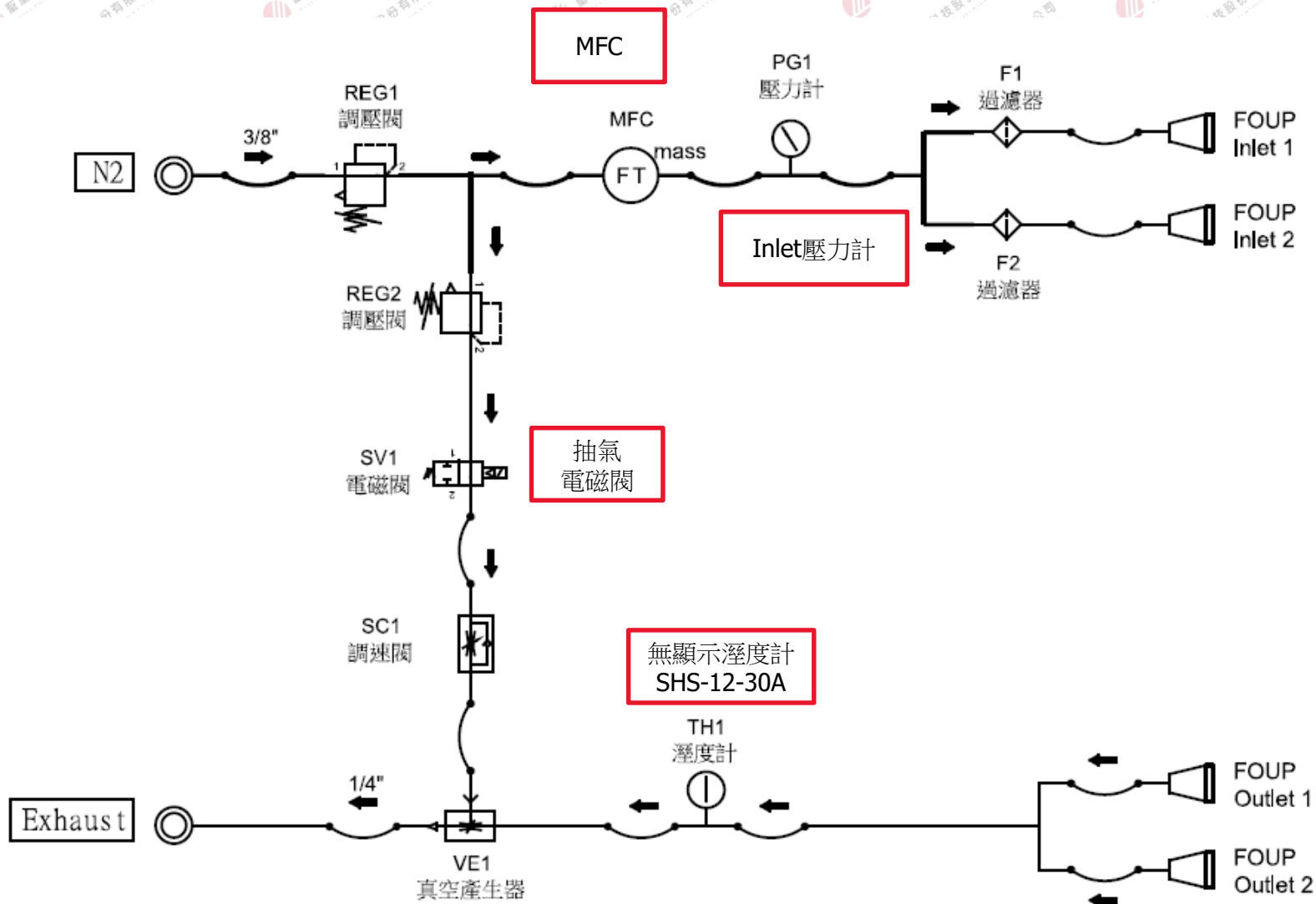
美光 EQ_LP 電氣路元件選型

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美光 EQ_LP 氣路圖



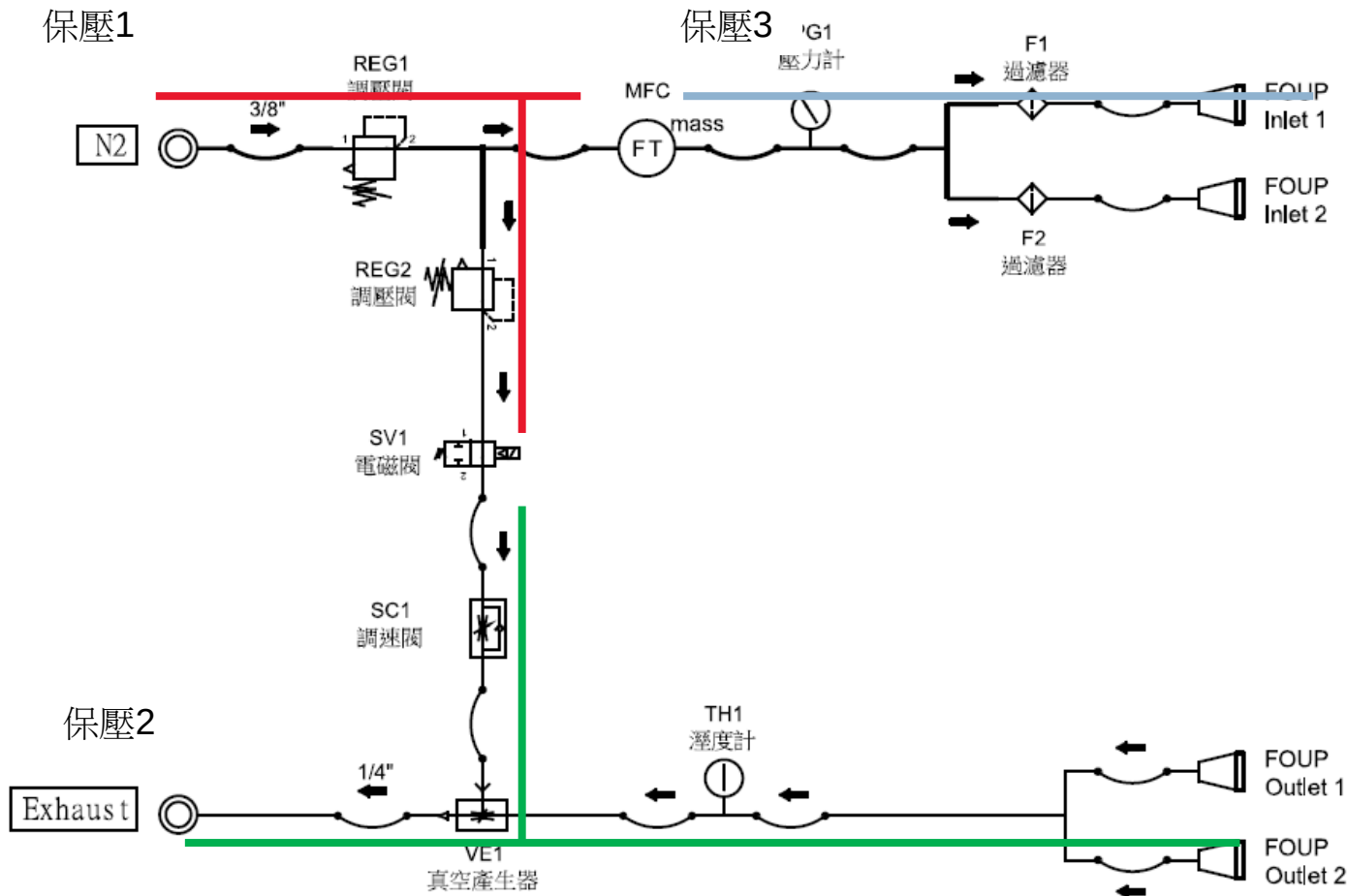


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美光 EQ_LP 保壓測試

保壓1

保壓3





機台檢查測試項目1

配製盤電路	各線路檢查確認有無短路	
	電壓AC 110 V 輸入	
	電壓DC 24 V輸出	
	PLC I/O接點燈號檢測	
	PLC DIP switch to RUN1	
	E84、PLC、RFID線路測試	
	箱體名牌、警告標示黏貼	
氣路	BOX管路漏氣檢測	
	Flow Rate 0~200L 測試	
	Exhaust抽氣流量檢測	
	A300 : 0~10L	
	電磁閥測試	
	PUMP、CDA流量檢測	
	數位流量開關功能	
	EMO開關、O2偵測器測試	
	Clamp 調壓閥壓力設定 ???	Standalone機台
	Clamp 氣路保壓	
	Clamp 速度調整(使用電控提供程式)	



機台檢查測試項目2

Load Port Plate	Load Port管路漏氣檢測
	Sensor 訊號檢測
	SEP噴嘴量測 (無SEP · 可跳過此項目)
	氣動閥測試
Pc 規格	CPU: I7
	RPM:32.00 GB
	HDD:SSD 250G*2
	系統類型:WIN 10 (64位元)
	病毒掃描:Office Scan
	GUI字串移除 (RFID Carried·····)



各Port Modbus元件參數設定表

Modbus 元件 Address站號表				
	PLC	壓力計	溫溼度計	MFC
Port 1	NA	21	41	61
Port 2	2	22	42	62
Port 3	3	23	43	63
Port 4	4	24	44	64

通訊參數設定

通訊格式 Rs-485

BaudRate : 38400

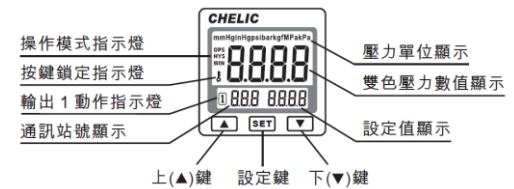
WordLength : 8

StopBits : 1

Parity : **EVEN**

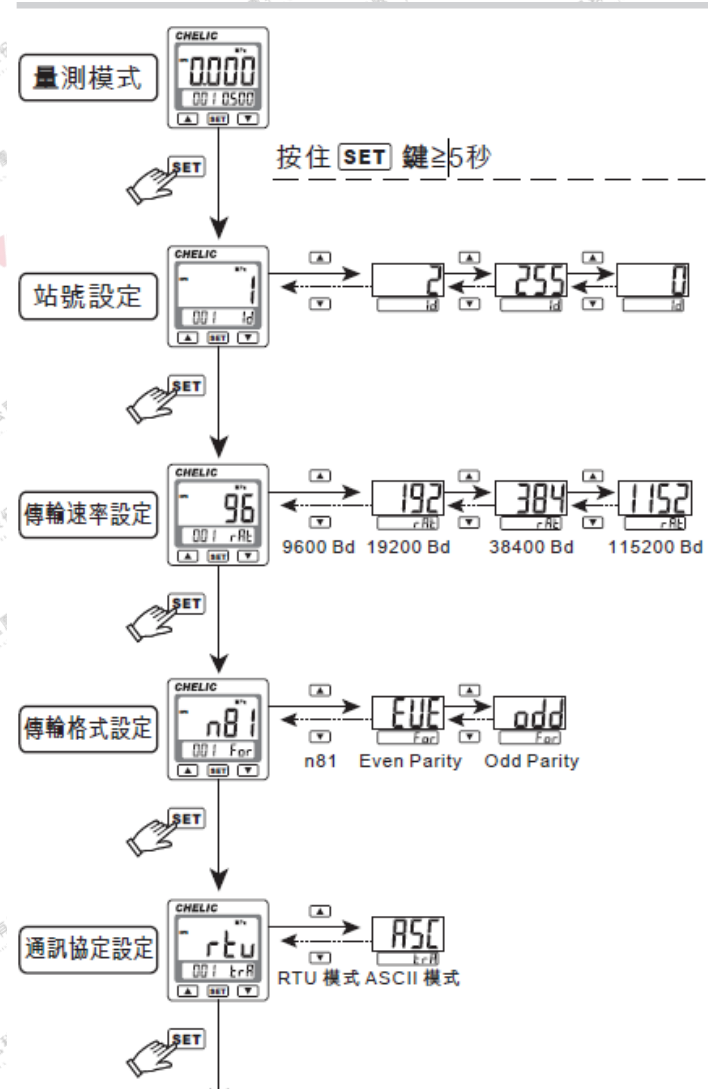


壓力計設定1



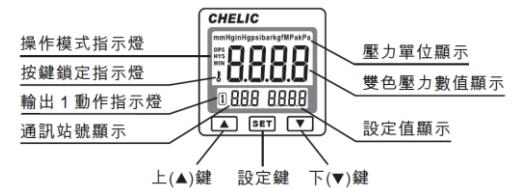
- Step1. Modbus通訊參數設定
- 在常態下按壓流量計按鈕“S”5 秒以上進入進階參數設定狀態。
- 顯示設定為1000

通訊參數設定
通訊格式 Rs-485
BaudRate : 38400
WordLength : 8
StopBits : 1
Parity : ENEN



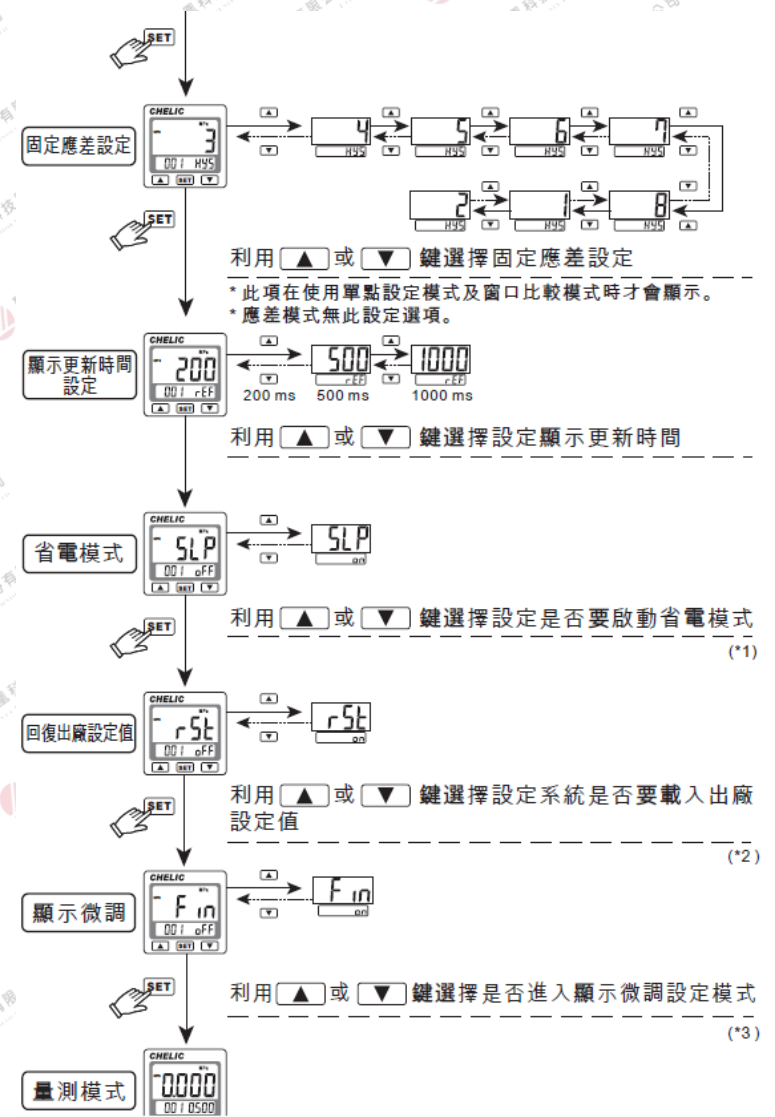


壓力計設定1-2

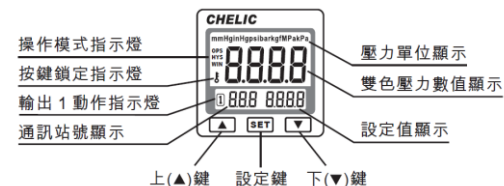


- Step1. Modbus通訊參數設定
- 在常態下按壓流量計按鈕“S”5 秒以上進入進階參數設定狀態。
- 顯示設定為1000

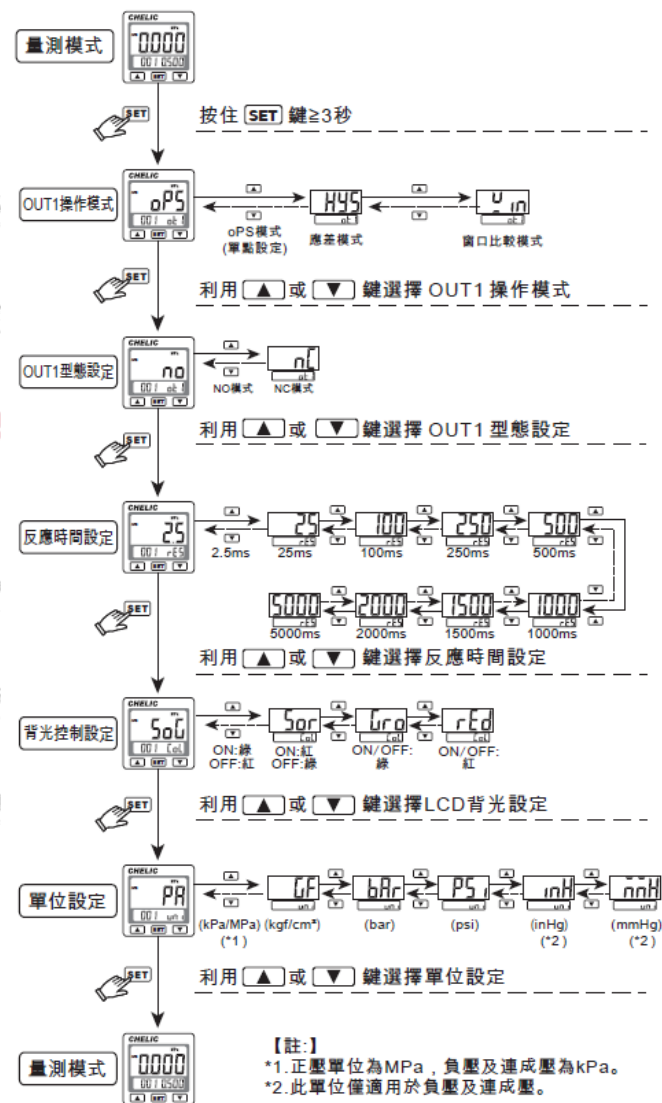
通訊參數設定
通訊格式 Rs-485
BaudRate : 38400
WordLength : 8
StopBits : 1
Parity : EVEN



壓力計設定2



- Step1. 基本參數設定
- 在常態下按壓流量計按鈕"**S**"3 秒以上進入基本參數設定狀態。
- 反應時間設定**1000**
- 單位設定**Mpa**





- MFC Modbus通訊參數設定
- STEP1. 打開Bronkhorst 軟體 FlowDDe

 FlowDDE V4.84

File Communication FLOW-BUS Server Info

Interface: not active

COM9 38400,n,8,1,-

Errors: 0

Msg/sec: 0

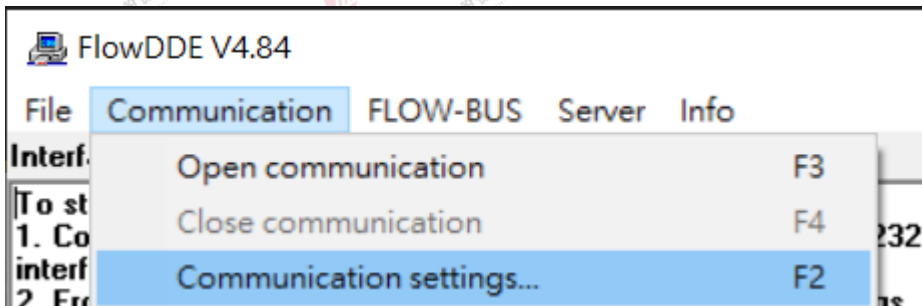
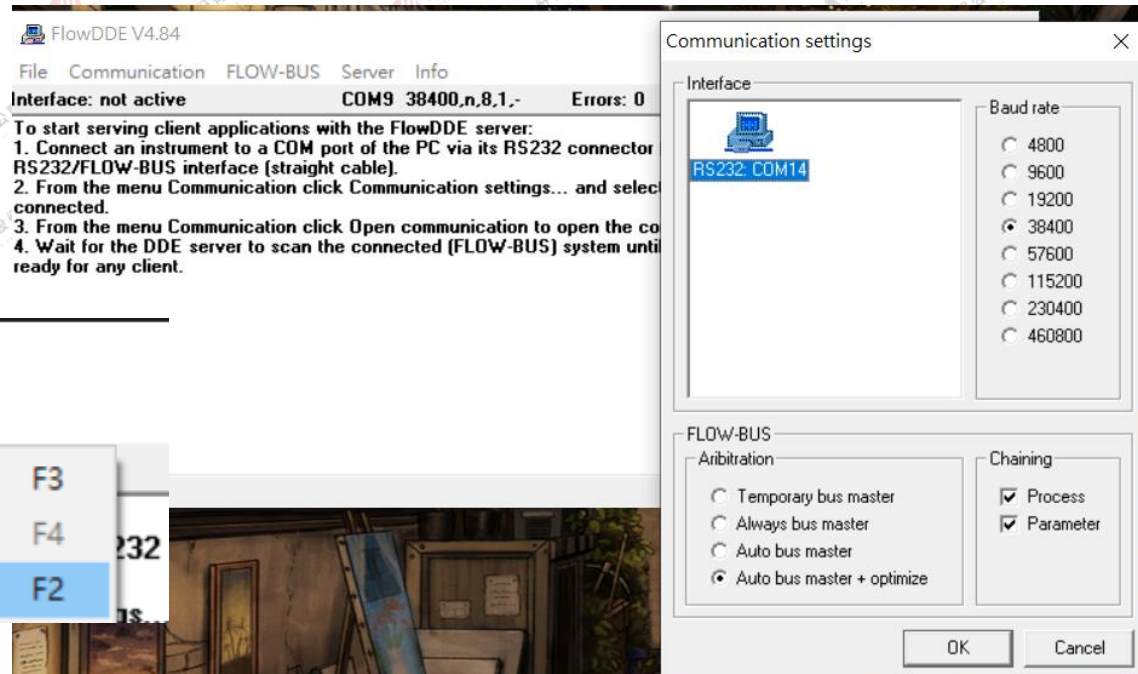
Msg: 0

To start serving client applications with the FlowDDE server:

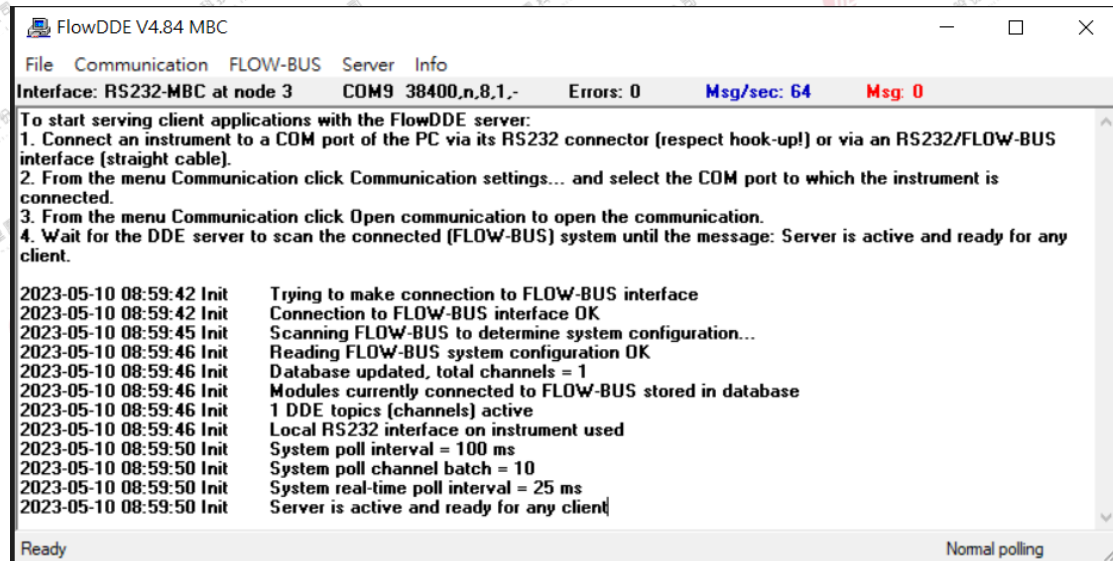
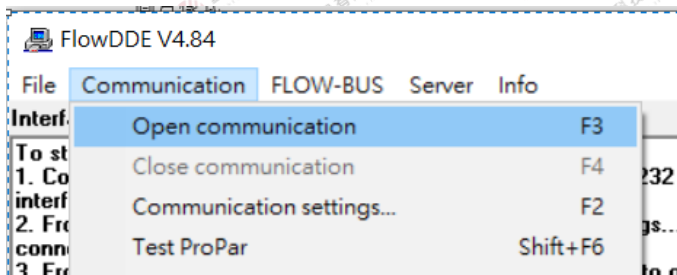
1. Connect an instrument to a COM port of the PC via its RS232 connector (respect hook-up!) or via an RS232/FLOW-BUS interface (straight cable).
2. From the menu Communication click Communication settings... and select the COM port to which the instrument is connected.
3. From the menu Communication click Open communication to open the communication.
4. Wait for the DDE server to scan the connected (FLOW-BUS) system until the message: Server is active and ready for any client.



- MFC Modbus通訊參數設定
- STEP2. 點選Communication 頁簽內的Communication settings。點選連線Com 在點擊OK即可。

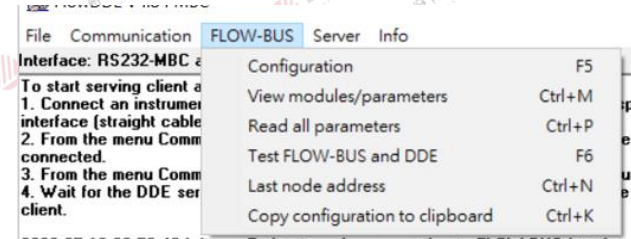


- MFC Modbus通訊參數設定
- STEP3. 點選Communication 頁簽內的Open Communication。開始進行連線，如果中途跳出彈跳視窗一律點選Yes即可。



■ MFC Modbus通訊參數設定

■ STEP4. 點選Flow-Bus業簽的Test Flow-BUS and DDE。進行通訊設定



參數保護機制，要先設定成64才能寫入參數。
設定完後需要再設定回82

Test form FlowDDE

Channel:	Parameter:	F5	Read value:	Write value:	F6
Ch: 1, DMFC, node 3, process 1	81: Global modify	Read	1	15	Write
Ch: 1, DMFC, node 3, process 1	7: Initreset	Read	82	82	Write
Ch: 1, DMFC, node 3, process 1	86: IO status	Read	15	15	Write
Ch: 1, DMFC, node 3, process 1	199: Fieldbus1 address	Read	63	64	Write
Ch: 1, DMFC, node 3, process 1	201: Fieldbus1 baudrate	Read	38400	38400	Write
Ch: 1, DMFC, node 3, process 1	335: Fieldbus1 parity	Read	2	2	Write

Channel:	Parameter:	F7	Read value:	Write value:	F8
Ch: 1, DMFC, node 3, process 1	0: poll parameters	Read	0		Write

☐ Disable standard polling

Close

通訊參數設定
站別
通訊速率



- STEP1. 安裝Festo 軟體，並安裝其相對應驅動（軟體部分可參閱FAS連線VEMD步驟PDF檔案
- 路徑如下所示
- \\192.168.1.98\專案資料\專案程式&說明書&電路圖面\專案程式\軟韌體_主程式\FESTO MFC 設定軟體)



- STEP2. 點選電控事先設定好的FESTO MFC 專案檔案。(如有問題請參閱 安專軟體內的 PDF檔挑選對應的元件型號進行設定。)





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FESTO MFC設定 3

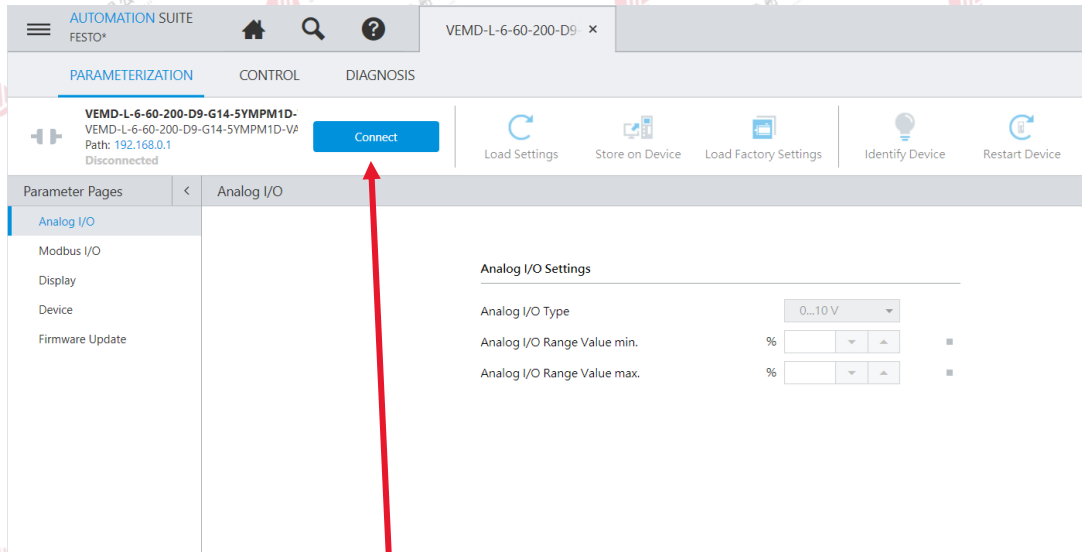
- STEP3. 點開後，會自動開始Festo 軟體，畫面開啟後請點擊正中央VEMD-L-6-60-200元件。



VEMD-
L-6-60-200-...



- **STEP4.** 等待一陣子後會跳轉至下一頁面。
- 初始MFC 網域皆為**192.168.0.1**，請準備一條正常網路線並將自己電腦網卡介面設定成同網域



- 設定好後請接上網路線，並點**Connect**按鈕
- 通常連線有可能花費**3-5**分鐘請耐心等待。



最後一碼不能設定為**1**或**0**，請設定其他數字



- STEP5. 等待一陣子後會跳轉至下一頁面。
- 連線成功後，按鈕會呈現Disconnet，左邊會打勾顯示連線Connected

The screenshot displays the 'AUTOMATION SUITE' interface for a FESTO device. The top navigation bar includes a home icon, a search icon, and a help icon, with the device ID 'VEMD-L-6-60-2' on the right. Below this, the 'PARAMETERIZATION' tab is active, alongside 'CONTROL' and 'DIAGNOSIS' tabs. The main area shows a green checkmark icon, the device model 'VEMD-L-6-60-200-D9-G14-5YMPM1D-V', and the IP address 'Path: 192.168.0.1'. A 'Disconnect' button is visible. To the right, a 'Load Setti' button with a circular arrow icon is present. On the left, a 'Parameter Pages' sidebar lists 'Analog I/O' (selected), 'Modbus I/O', 'Display', 'Device', and 'Firmware Update'. The main content area shows 'Analog I/O' with three sub-items, each labeled 'Analog I/O'.



- STEP6. 點選Modbus I/O，確認左側的通訊設定是否正確(請參閱上面的站別通訊設定表。)
- 注意，如果修改的內容有站別。請點選Store on Device後再按Restart Device結束連線，再重新連線進入此頁面。確認站別設定有正確寫入)

The screenshot displays the Festo MFC configuration software interface. At the top, a status bar shows a green checkmark, the device ID 'VEMD-L-6-60-200-D9-G14-5YMPM1D-V', and the path '192.168.0.1' with a 'Connected' status. Below this are buttons for 'Disconnect', 'Load Settings', 'Store on Device' (circled in blue), 'Load Factory Settings', 'Identify Device', and 'Restart Device' (circled in blue). The main area is divided into a left sidebar and a right content pane. The sidebar, titled 'Parameter Pages', lists 'Analog I/O', 'Modbus I/O' (highlighted with a red box), 'Display', 'Device', and 'Firmware Update'. The right pane, titled 'Modbus I/O Settings', contains a warning message: 'Changing the Modbus address will need a restart of the device.' Below the warning are four settings: 'Modbus Address' (set to 61), 'Baudrate' (set to 38400 Bd), 'Stopbits' (set to 1 bit), and 'Parity' (set to Even). The entire settings pane is enclosed in a red rectangular box.



- STEP7. 確認通訊與站別都正確後，點選Control。確認MFC 內容如下表所示

PARAMETERIZATION **CONTROL** DIAGNOSIS

✓ VEMD-L-6-60-200-D9-G14-SYMPM1D-
VEMD-L-6-60-200-D9-G14-SYMPM1D-VA
Path: 192.168.0.1
Connected

Disconnect

Load Settings Store on Device Load Factory Settings Identify Device

Control Pages < Control

Control

Control Settings

Actual Flow Filter	ms	100	▼ ▲	■
Setpoint Filter	ms	100	▼ ▲	■
Maximum Flow	%	100	▼ ▲	■
Setpoint Source		Digital	▼	
Control Type		Flow	▼	

Real Time Control

ⓘ This value will be not stored on the device permanently.

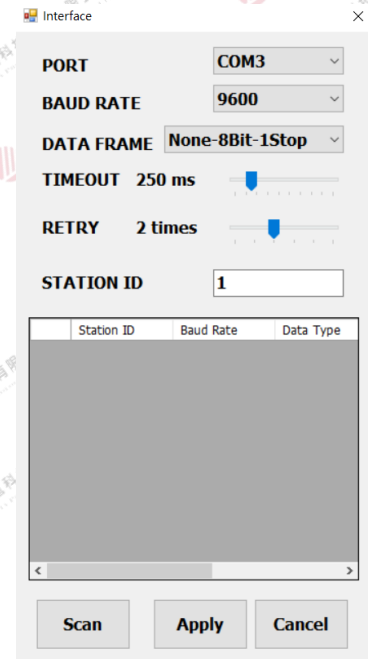
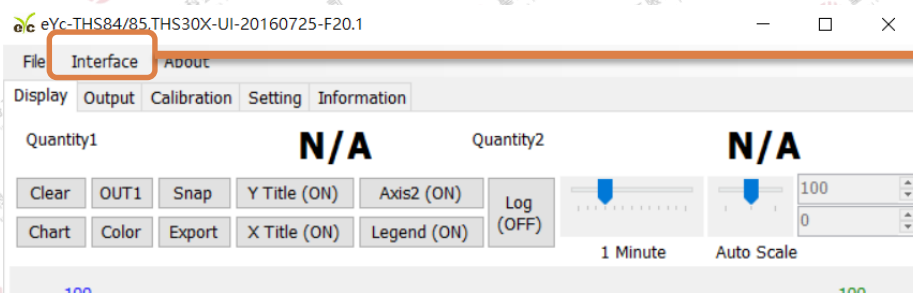
Digital Setpoint % 0.00 ☒

- STEP8. 完成設定後，將MFC斷線再重新連線。再次確認所有參數是否都正確即可。



溼度計設定1

- Step1. 運用Rs 485與溼度計進行連線。
- 開啟溼度計程式，點選Interface表單->Config
- 跳出通訊設定頁面後，依據連接Com點進行連接，通訊參數初始為9600.n.8.1，Station ID(站別)1
後續有修正須依據修正參數進行設定。





溼度計設定2

- Step2. 依據出機的業主以及廠區輸入相關濕度校正參數，校正參數表依據RD於網路提供的最新版本為主。
- 路徑：
\\192.168.1.98\專案資料\設備參數 區

版次	設定項目	原廠設定值 設定方式	參數表						使用客戶											
			必要參數1	必要參數2	必要參數3	必要參數4	必要參數5	必要參數6												
V1.02 高低濕	目前偵測值 (Hunit%)	當原廠參數偵測值或顯示值其中一個在設定參數範圍內時，移除該原廠參數。其他保留	45	25	20	8	3	-2	南亞	TSMC-F18	TSMC-F15A	TSMC-F15E	村田	大福	福懋	PSMC	美光日本 (F15)			
	目前顯示值 (Hstd%)		45	40	30	3.5	0.5	0.5												
V3.03 低濕	目前偵測值 (Hunit%)	當原廠參數偵測值或顯示值其中一個在設定參數範圍內時，移除該原廠參數。其他保留	8	3	-2				TSMC-F12B	TSMC-F14A	TSMC-F14B	TSMC-F16	SYZ (其他)	華邦	其他	日月光	大陸 (聖育澤)	UMC-聯華電子	美光后里 (F16)	美光林口 (F11)
	目前顯示值 (Hstd%)		3.5	0.3	0.3															

eYc-THS84/85,THS30X-UI-20160725-F20.1 Station 1*

File Interface About

Display Output **Calibration** Setting Information

Temperature

UPPER POINT

LOWER POINT

OFFSET POINT

RECALL FACTORY CALIB

	Tunit [°C]	Tstd [°C]
▶		

WRITE

READ

CLEAR

RECALL

Relative Humidity

UPPER POINT

LOWER POINT

OFFSET POINT

RECALL FACTORY CALIB

	Hunit [%]	Hstd [%]
▶	61.2635	60.05
	80.6472	80
	45	45
	25	40
	20	30
	8	3.5
	3	0.5
	-2	0.5

WRITE

READ

CLEAR

RECALL

Open Port. Read successful



- Step3.設定Modbus相關通訊設定參數(會牽扯到下次連線時的通訊參數，請多注意。)，點選**Setting** 進入設定頁面。
- 設定參數請參閱下列內容，或參閱**Modbus**元件表
- **Station ID (站號):**依據**Port Number**設定
- **Baud Rate (通訊):**38400
- **Data Frame :** E-8-1
- **Response Time :** 5
- 參數設定完成後點選**Apply**
- 點選**Apply**後需重新連線
連線設定須依據剛剛通訊參數設定輸入

eYc-THS84/85,THS30X-UI-20160725-F20.1 Station 41*

File Interface About

Display Output Calibration Setting Information

Environment

Air Pressure (mBar) 1013.25

Modbus Protocol

Station ID 41

Baud Rate 38400

Data Frame None-8Bit-1Stop

Response Time (ms) 5

Temperature Unit

☒ °C
☐ °F

Test Count: Test Result
Write Error:
Read Error:
Data Error:

Echo Test (OFF) Reset Counter

Apply Read

Read OUT1 Config, Read successful



美光 EQ_LP 抽氣流量設定

- 1.將調壓閥設定為80psi後，進行抽氣作業。(使用Auto Purge來進行調整)
- 2.調整抽氣端調節閥(0.3mpa)，比對流量計數值旋轉調節閥，將流量設定為3.3 L/min。
。調整完成後請反覆充氣3~5次確認流量是否正確及穩定。

請在有FOUP狀態下完成上述調整，避免放上FOUP後有些微誤差。

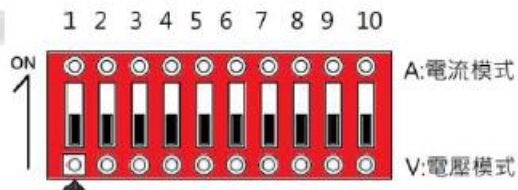


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美光 EQ_LP 類比指撥(上板)

■ 指撥設定請參閱下表

- 指撥開關定義(SW1_10P):類比輸入模式切換

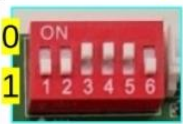


Modbus類型機台，上層指撥於電流或電壓模式不影響。
統一將指撥撥至OFF位置



■ 指撥設定請參閱下表

指撥號碼	1	2	3	4	5	6
指撥方向	0	1	0	0	0	0

DIP-Switch功能	
	
編號	數值與功能
1、2	00: 預留 01: 預留 10: 預留 11: 力積電
3、4	00: Bronkhorst 01: Festo 10: 預留 11: 預留
5	0: 115200 (RS232) 1: 9600 (RS232)
6	0: RS232 通訊 1: TCP 通訊

3、4腳請依據使用的MFC來設定

● 功能為切換韌體流程與通訊設定功能



該圖為指撥定義說明請根據現場需求設定

■ N2P 通訊Com點請依據安裝位置設定，通訊速率改為115200

```
</NVTtree>
```

```
<NVTtree name="N2P.PLC1">
  <NV name ="COMPortSetting"
  <NV name ="LinkTimeout"
  <NV name ="CmdDelayTime"
  <NV name ="CmdReplyTimeout"
```

```
value ="COM1,115200,n,8,1" />
value ="10" desc="link timeout[5~20, default= 10, seconds]/>
value ="0.2" desc="delay time for sending next command seconds [0~0.5, default=0.2, seconds]"/>
value ="5" desc ="plc command reply timeout [0.5~, default=1, seconds]" />
```




Port Parameter Settings					
Parameter Settings					
Param	Spec	Value	Unit	New Value	Update
HIGH_FLOW_MODE_HI_FLOW_LIMIT	$0 \leq \text{value} \leq 200$	0	l/m		set
HIGH_FLOW_MODE_LO_FLOW_LIMIT	$0 \leq \text{value} \leq 200$	0	l/m		set
HIGH_FLOW_MODE_HI_PRESSURE_LIMIT	$0 \leq \text{value} < 1$	0	Mpa		set
HIGH_FLOW_MODE_LO_PRESSURE_LIMIT	$0 \leq \text{value} < 1$	0	Mpa		set
LOW_FLOW_MODE_HI_FLOW_LIMIT	$0 \leq \text{value} \leq 200$	0	l/m		set
LOW_FLOW_MODE_LO_FLOW_LIMIT	$0 \leq \text{value} \leq 200$	0	l/m		set
LOW_FLOW_MODE_HI_PRESSURE_LIMIT	$0 \leq \text{value} < 1$	0	Mpa		set
LOW_FLOW_MODE_LO_PRESSURE_LIMIT	$0 \leq \text{value} < 1$	0	Mpa		set
DOOR_OPEN_DELAY_TIME	$0 \leq \text{value} \leq 20$	2	sec		set
IdlePurgeFlow	$4 \leq \text{value} \leq 200$	40	l/min		set
IdlePurgeTime	$2 \leq \text{value} \leq 30$	10	sec		set
IdlePurgeIntervalTime	$60 \leq \text{value} \leq 3600$	3600	sec		set
FoupPressingTime	$0 \leq \text{value} \leq 99$	1	sec		set
FoupSensor	$1 \leq \text{value} \leq 3$	3	na		set

依據LP盤Sensor數量設定



- Idle Purge 設定與測試
- 於參數頁面中，填入相關數值並SET。
- 當機台在無Alarm、Purge且FOUP沒在LP盤上時，當時間到達會進行Idle Purge動作。

Port Parameter Settings					
Parameter Settings					
Param	Spec	Value	Unit	New Value	Update
HIGH_FLOW_MODE_HI_FLOW_LIMIT	0 <= value <= 200	0	l/m		set
HIGH_FLOW_MODE_LO_FLOW_LIMIT	0 <= value <= 200	0	l/m		set
HIGH_FLOW_MODE_HI_PRESSURE_LIMIT	0 <= value < 1	0	Mpa		set
HIGH_FLOW_MODE_LO_PRESSURE_LIMIT	0 <= value < 1	0	Mpa		set
LOW_FLOW_MODE_HI_FLOW_LIMIT	0 <= value <= 200	0	l/m		set
LOW_FLOW_MODE_LO_FLOW_LIMIT	0 <= value <= 200	0	l/m		set
LOW_FLOW_MODE_HI_PRESSURE_LIMIT	0 <= value < 1	0	Mpa		set
LOW_FLOW_MODE_LO_PRESSURE_LIMIT	0 <= value < 1	0	Mpa		set
DOOR_OPEN_DELAY_TIME	0 <= value <= 20	2	sec		set
IdlePurgeFlow	4 <= value <= 200	40	l/min		set
IdlePurgeTime	2 <= value <= 30	10	sec		set
IdlePurgeIntervalTime	60 <= value <= 3600	3600	sec		set
FoupPressingTime	0 <= value <= 99	1	sec		set
FoupSensor	1 <= value <= 3	3	na		set

Idle purge充氣流量

Idle purge充氣時間

Idle purge間隔時間



- Alarm測試
- 於GUI頁面上設定相關參數，並進行Alarm測試。(參數若設定為0，則Bypass該Alarm。)
- Alarm測試方法為，當N2機台開始Purge後30秒進行判斷壓力流量數值是否在範圍內，若高於或低於發出對應Alarm Code與Alarm 訊息。
- HIGH Flow Mode flow USL 大流量模式流量上限(USL=Upper Spec Limit)
- HIGH Flow Mode flow LSL 大流量模式流量下限(LSL=Lower Spec Limit)
- HIGH Flow Mode flow Pressure USL 大流量模式壓力上限
- HIGH Flow Mode flow Pressure LSL 大流量模式壓力下限
- Low Flow Mode flow USL小流量模式流量上限
- Low Flow Mode flow LSL小流量模式流量下限
- Low Flow Mode flow Pressure USL小流量模式壓力上限
- Low Flow Mode flow Pressure LSL小流量模式壓力下限



- Door Open Delay Time 測試
- 設定該參數數值，假定設置5秒。
- 當**Door Open Sensor** 狀態從 1(On) -> 0(Off)，5秒後才會重大流量切成小流量。
- 若設定為0 則立馬切換大小流量變化。
- ◎Door Open Sensor 狀態從 0(Off) -> 1(On)不受該參數影響。

Port Parameter Settings					
Parameter Settings					
Param	Spec	Value	Unit	New Value	Update
HIGH_FLOW_MODE_HI_FLOW_LIMIT	0 <= value <= 200	0	l/m		set
HIGH_FLOW_MODE_LO_FLOW_LIMIT	0 <= value <= 200	0	l/m		set
HIGH_FLOW_MODE_HI_PRESSURE_LIMIT	0 <= value < 1	0	Mpa		set
HIGH_FLOW_MODE_LO_PRESSURE_LIMIT	0 <= value < 1	0	Mpa		set
LOW_FLOW_MODE_HI_FLOW_LIMIT	0 <= value <= 200	0	l/m		set
LOW_FLOW_MODE_LO_FLOW_LIMIT	0 <= value <= 200	0	l/m		set
LOW_FLOW_MODE_HI_PRESSURE_LIMIT	0 <= value < 1	0	Mpa		set
LOW_FLOW_MODE_LO_PRESSURE_LIMIT	0 <= value < 1	0	Mpa		set
DOOR_OPEN_DELAY_TIME	0 <= value <= 20	2	sec		set
IdlePurgeFlow	4 <= value <= 200	40	l/min		set
IdlePurgeTime	2 <= value <= 30	10	sec		set
IdlePurgeIntervalTime	60 <= value <= 3600	3600	sec		set
FoupPressingTime	0 <= value <= 99	1	sec		set
FoupSensor	1 <= value <= 3	3	na		set



- LP Sensor 測試
- FoupSensor可設定Sensor數量(請依據盤上數量進行設定，並接上Sensor)
- 當設定1時，FOUP Sensor 異常判斷Alarm 不有任何作用。
- 當設定2時，FOUP Sensor 異常判斷Alarm 啟動，當X00、X01 Sensor狀態不同超過 FOUP Pressing Time 設定參數時，觸發Alarm。
- 當設定3時，FOUP Sensor 異常判斷Alarm 啟動，當X00、X01 Sensor狀態不同超過 FOUP Pressing Time 設定參數時，觸發Alarm。
- 現在判斷有無**FOUP**
X00、X01、X06都需要按壓狀態
- **Ex:**若只有按壓**X00、X01**，機台判定無**FOUP**
- **◎Sensor無接狀態下，認定該點位有被按壓。**

X01、X02異常Delay Time
LP Sensor數量

Port Parameter Settings						
Parameter Settings						
Param	Spec	Value	Unit	New Value	Update	
HIGH_FLOW_MODE_HI_FLOW_LIMIT	0 <= value <= 200	0	l/m		set	
HIGH_FLOW_MODE_LO_FLOW_LIMIT	0 <= value <= 200	0	l/m		set	
HIGH_FLOW_MODE_HI_PRESSURE_LIMIT	0 <= value < 1	0	Mpa		set	
HIGH_FLOW_MODE_LO_PRESSURE_LIMIT	0 <= value < 1	0	Mpa		set	
LOW_FLOW_MODE_HI_FLOW_LIMIT	0 <= value <= 200	0	l/m		set	
LOW_FLOW_MODE_LO_FLOW_LIMIT	0 <= value <= 200	0	l/m		set	
LOW_FLOW_MODE_HI_PRESSURE_LIMIT	0 <= value < 1	0	Mpa		set	
LOW_FLOW_MODE_LO_PRESSURE_LIMIT	0 <= value < 1	0	Mpa		set	
DOOR_OPEN_DELAY_TIME	0 <= value <= 20	2	sec		set	
IdlePurgeFlow	4 <= value <= 200	40	l/min		set	
IdlePurgeTime	2 <= value <= 30	10	sec		set	
IdlePurgeIntervalTime	60 <= value <= 3600	3600	sec		set	
FoupPressingTime	0 <= value <= 99	1	sec		set	
FoupSensor	1 <= value <= 3	3	na		set	



- Auto Purge 模式測試
- 大小流量充氣切換測試:
- 將FOUP門打開並放置LP盤上，按下Stop Purge 進行Purge 動作。
 - a. 按壓Door Open sensor確認大小流量是否正常切換。
 - b. 確認流量切換後是否穩定且抽氣與充氣流量數值有無跑掉，如有誤差請修正。
 - c. 反覆切換5次，每2分鐘進行切換確認抽氣與充氣數值穩定度。
 - d. 拿走FOUP後是否，Purge動作是否停止。



- **Maintain Purge 模式測試**
- 小流量充氣測試:
- 將FOUP放置LP盤上進行 Low Flow Purge 測試 (抽氣3.3L/min 充氣40L/min)，記錄每一分鐘濕度數值(手持式與無顯示溼度計)。當十分鐘後濕度是否低於1%，如沒有請確認相關氣路是否有漏氣現象發生或FOUP沒放置好。
- **測試前請先確認初始濕度為40~50%。**
- 大流量充氣測試:
- 將FOUP門打開，放置LP盤上High Flow Purge 測試 (抽氣3.3L/min 充氣160L/min)，噴氣3分鐘確認數值流量是否有跳動現象，如有跳動請在調整相關調節閥。



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GUI N2P 設定 1對4 -1

```
<!--by equipment -->
<NV name="PLCCount" value="1" desc="total PLC count" />
<NV name="PortCount" value="4" desc="total Port count" />

<!--<NVTree name="FanController">
  <NV name="PlcSettings" value="Type=RTU;Station=1;COMPort=COM6,115200,n,8,1" />
</NVTree-->

</NVTree>

<NVTree name="N2P.PLC1">
  <NV name="COMPortSetting" value="COM1,115200,n,8,1" />
  <NV name="LinkTimeout" value="10" desc="link timeout[5~20, default= 10, seconds]"/>
  <NV name="CmdDelayTime" value="0.2" desc="delay time for sending next command seconds [0~0.5, default=0.2, seconds]"/>
  <NV name="CmdReplyTimeout" value="5" desc="plc command reply timeout [0.5~, default=1, seconds]" />
</NVTree>
```



```
<NVTTree name="N2P.PORT 1" >
  <NV name="PLC" value="PLC1" />
  <NV name="PLCPortNo" value="1" />
  <NV name="PortNo" value="1" desc="Host port No" />
  <NV name="DisplayName" value="PORT 1" />

  <NV name="SupportMFCType" value="true" desc="default false" />
  <!--MFC type, 1=Bronkhorst, 2=Festo / -->
  <NV name="MFCType" value="1" desc="default 1" />

</NVTTree>

<NVTTree name="N2P.PORT 2" >
  <NV name="PLC" value="PLC1" />
  <NV name="PLCPortNo" value="2" />
  <NV name="PortNo" value="2" desc="Host port No" />
  <NV name="DisplayName" value="PORT 2" />

  <NV name="SupportMFCType" value="true" desc="default false" />
  <!--MFC type, 1=Bronkhorst, 2=Festo / -->
  <NV name="MFCType" value="1" desc="default 1" />

</NVTTree>

<NVTTree name="N2P.PORT 3" >
  <NV name="PLC" value="PLC1" />
  <NV name="PLCPortNo" value="3" />
  <NV name="PortNo" value="3" desc="Host port No" />
  <NV name="DisplayName" value="PORT 3" />

  <NV name="SupportMFCType" value="true" desc="default false" />
  <!--MFC type, 1=Bronkhorst, 2=Festo / -->
  <NV name="MFCType" value="1" desc="default 1" />

</NVTTree>

<NVTTree name="N2P.PORT 4" >
  <NV name="PLC" value="PLC1" />
  <NV name="PLCPortNo" value="4" />
  <NV name="PortNo" value="4" desc="Host port No" />
  <NV name="DisplayName" value="PORT 4" />

  <NV name="SupportMFCType" value="true" desc="default false" />
  <!--MFC type, 1=Bronkhorst, 2=Festo / -->
  <NV name="MFCType" value="1" desc="default 1" />

</NVTTree>
```



PLC 1對1

```
<!--by equipment -->
<NV name="PLCCount" value="4" desc="total PLC count" />
<NV name="PortCount" value="4" desc="total Port count" />

<!--<NVTre name ="FanController">
  <NV name ="PlcSettings" value="Type=RTU;Station=1;COMPort=COM6,115200,n,8,1" />
</NVTre-->

</NVTre>

<NVTre name="N2P.PLC1">
  <NV name ="COMPortSetting" value ="COM1,9600,n,8,1" />
  <NV name ="LinkTimeout" value ="10" desc="link timeout[5~20, default= 10, seconds]/>
  <NV name ="CmdDelayTime" value ="0.2" desc="delay time for sending next command seconds [0~0.5, default=0.2, seconds]"/>
  <NV name ="CmdReplyTimeout" value ="5" desc ="plc command reply timeout [0.5~, default=1, seconds]" />

</NVTre>

<NVTre name="N2P.PLC2">
  <NV name ="COMPortSetting" value ="COM2,9600,n,8,1" />
  <NV name ="LinkTimeout" value ="10" desc="link timeout[5~20, default= 10, seconds]/>
  <NV name ="CmdDelayTime" value ="0.2" desc="delay time for sending next command seconds [0~0.5, default=0.2, seconds]"/>
  <NV name ="CmdReplyTimeout" value ="5" desc ="plc command reply timeout [0.5~, default=1, seconds]" />

</NVTre>

<NVTre name="N2P.PLC3">
  <NV name ="COMPortSetting" value ="COM3,9600,n,8,1" />
  <NV name ="LinkTimeout" value ="10" desc="link timeout[5~20, default= 10, seconds]/>
  <NV name ="CmdDelayTime" value ="0.2" desc="delay time for sending next command seconds [0~0.5, default=0.2, seconds]"/>
  <NV name ="CmdReplyTimeout" value ="5" desc ="plc command reply timeout [0.5~, default=1, seconds]" />

</NVTre>

<NVTre name="N2P.PLC4">
  <NV name ="COMPortSetting" value ="COM4,9600,n,8,1" />
  <NV name ="LinkTimeout" value ="10" desc="link timeout[5~20, default= 10, seconds]/>
  <NV name ="CmdDelayTime" value ="0.2" desc="delay time for sending next command seconds [0~0.5, default=0.2, seconds]"/>
  <NV name ="CmdReplyTimeout" value ="5" desc ="plc command reply timeout [0.5~, default=1, seconds]" />
```




```
<NVTTree name="N2P.PORT 1" >
  <NV name="PLC" value="PLC1" />
  <NV name="PLCPortNo" value="1" />
  <NV name="PortNo" value="1" desc="Host port No" />
  <NV name="DisplayName" value="PORT 1" />

  <NV name="SupportMFCType" value="true" desc="default false" />
  <!-- MFC type, 1=Bronkhorst, 2=Festo / -->
  <NV name="MFCType" value="1" desc="default 1" />

</NVTTree>

<NVTTree name="N2P.PORT 2" >
  <NV name="PLC" value="PLC2" />
  <NV name="PLCPortNo" value="1" />
  <NV name="PortNo" value="2" desc="Host port No" />
  <NV name="DisplayName" value="PORT 2" />

  <NV name="SupportMFCType" value="true" desc="default false" />
  <!-- MFC type, 1=Bronkhorst, 2=Festo / -->
  <NV name="MFCType" value="1" desc="default 1" />

</NVTTree>

<NVTTree name="N2P.PORT 3" >
  <NV name="PLC" value="PLC3" />
  <NV name="PLCPortNo" value="1" />
  <NV name="PortNo" value="3" desc="Host port No" />
  <NV name="DisplayName" value="PORT 3" />

  <NV name="SupportMFCType" value="true" desc="default false" />
  <!-- MFC type, 1=Bronkhorst, 2=Festo / -->
  <NV name="MFCType" value="1" desc="default 1" />

</NVTTree>

<NVTTree name="N2P.PORT 4" >
  <NV name="PLC" value="PLC4" />
  <NV name="PLCPortNo" value="1" />
  <NV name="PortNo" value="4" desc="Host port No" />
  <NV name="DisplayName" value="PORT 4" />

  <NV name="SupportMFCType" value="true" desc="default false" />
  <!-- MFC type, 1=Bronkhorst, 2=Festo / -->
  <NV name="MFCType" value="1" desc="default 1" />

</NVTTree>
```



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GUI N2P MFC Type設定

```
<NVTreename="N2P.PORT 1">
  <NVname="PLC" value="PLC1" />
  <NVname="PLCPortNo" value="1" />
  <NVname="PortNo" value="1" desc="Host port No" />
  <NVname="DisplayName" value="PORT 1" />

  <NVname="SupportMFCType" value="true" desc="default false" />
  <!--MFC type 1=Bronkhorst 2=Festo /-->
  <NVname="MFCType" value="1" desc="default 1" />

</NVTreename="N2P.PORT 2">
  <NVname="PLC" value="PLC1" />
  <NVname="PLCPortNo" value="2" />
  <NVname="PortNo" value="2" desc="Host port No" />
  <NVname="DisplayName" value="PORT 2" />

  <NVname="SupportMFCType" value="true" desc="default false" />
  <!--MFC type 1=Bronkhorst 2=Festo /-->
  <NVname="MFCType" value="1" desc="default 1" />

</NVTreename="N2P.PORT 3">
  <NVname="PLC" value="PLC1" />
  <NVname="PLCPortNo" value="3" />
  <NVname="PortNo" value="3" desc="Host port No" />
  <NVname="DisplayName" value="PORT 3" />

  <NVname="SupportMFCType" value="true" desc="default false" />
  <!--MFC type 1=Bronkhorst 2=Festo /-->
  <NVname="MFCType" value="1" desc="default 1" />

</NVTreename="N2P.PORT 4">
  <NVname="PLC" value="PLC1" />
  <NVname="PLCPortNo" value="4" />
  <NVname="PortNo" value="4" desc="Host port No" />
  <NVname="DisplayName" value="PORT 4" />

  <NVname="SupportMFCType" value="true" desc="default false" />
  <!--MFC type 1=Bronkhorst 2=Festo /-->
  <NVname="MFCType" value="1" desc="default 1" />

</NVTreename="N2P.PORT 5">
```



GUI 機台濕度log 存放位置

- 存放方式:資料庫
- 存放位置:D:\SPTI\N2pAddon\RunPDHistory (忘記了 可以查看N2P.xml)

```
<NVTTree name="PDHistory" >  
  <!-- use the PDCHistoryDir instead-->  
  <NV name="FileDir" value="D:\SPTI\N2pAddon\RunPDHistory" />  
  <NV name="KeepDays" value="90" />  
  <NV name="MaxSizeInMBytes" value="20" />  
</NVTTree>
```

本機 > DATA (D:) > SPTI > N2pAddon > RunPDHistory

名稱	修改日期	類型	大小
Backup	2022/10/18 下午 04:27	檔案資料夾	
Current	2022/10/31 下午 11:21	檔案資料夾	

- 資料庫超過20MB，會自動備份到Backup資料夾裡面，必須使用「ProcessDataQuery」收值軟體，正常都會放在D:\目錄底下

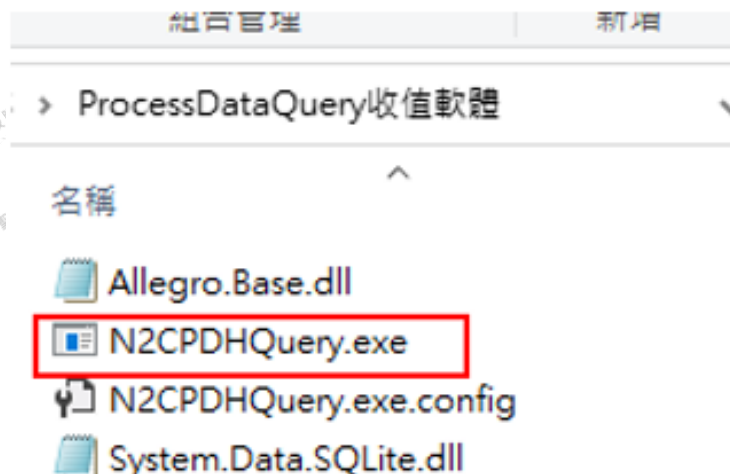


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Query Process data 工具SOP

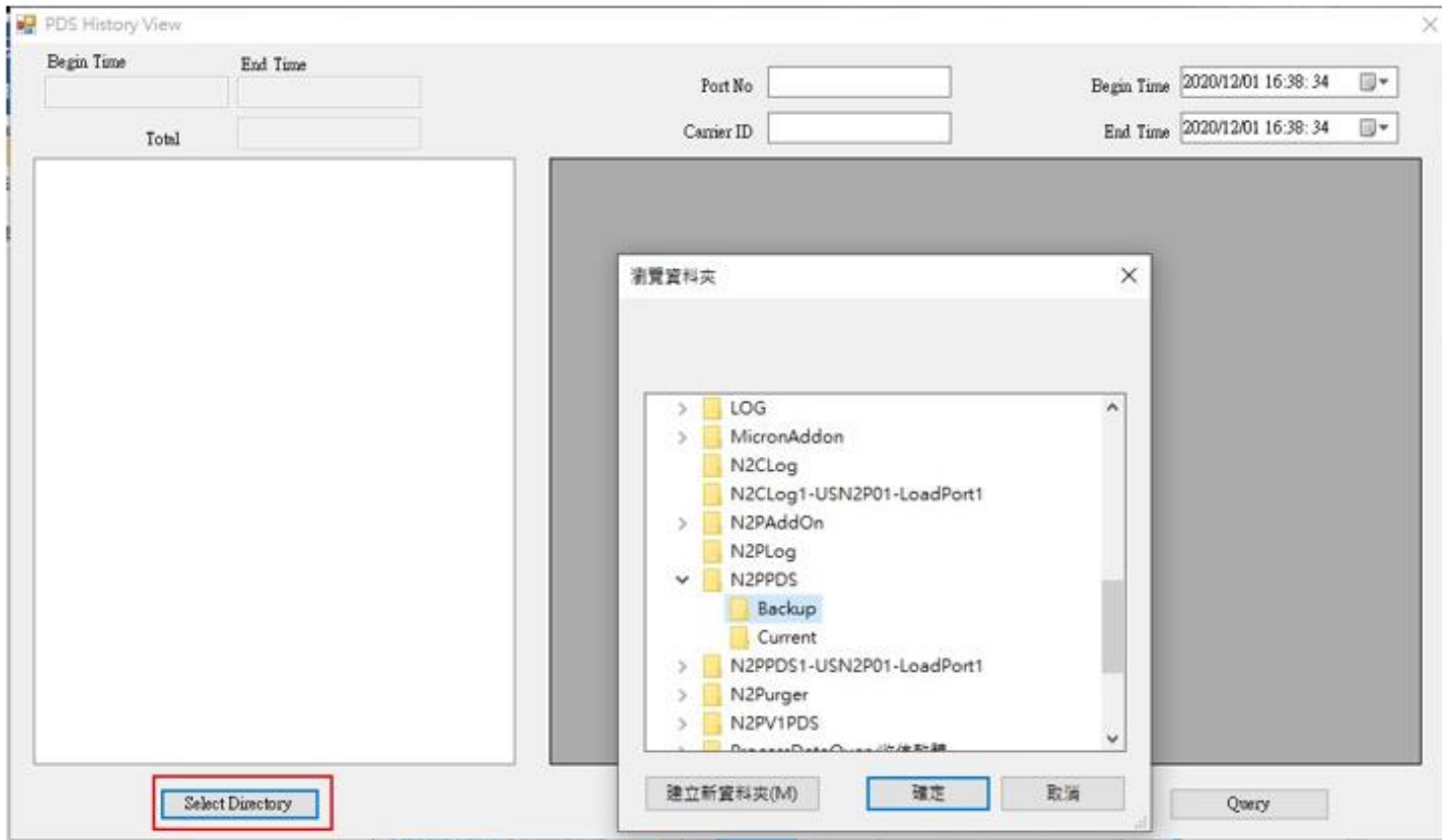
搜尋 Data 程式軟體 ←

Step1: 開啟『ProcessDataQuery』，執行 N2CPDHQuery.exe ←





Step2:點選『Select Directory』,選擇『D:\SPTI\N2pAddon\RunPDHistory\Backup』
資料夾←





Step3: 點選清單內檔案

PDS History View

Begin Time: 2015/03/09 10:36:54 End Time: 2015/03/09 15:45:05

Total: 66728

Port No: Carrier ID:

Begin Time: 2015/03/09 10:36:54 End Time: 2015/03/09 15:45:05

D:\N20PDS\SPPT\N20-PDS.db_20141213072127.bak
D:\N20PDS\SPPT\N20-PDS.db_20141215095149.bak
D:\N20PDS\SPPT\N20-PDS.db_20150309102935.bak
D:\N20PDS\SPPT\N20-PDS.db

點選清單內檔案，然後上面資料會顯示資料的起始時間與結束時間及Total數量。

TMST	PORT_NO	FOUP_ID	PPID	FLOWRATE	HUMIDITY
2015/03/09 15:4...	4	PFC854321	DDO	0	0.0
2015/03/09 15:4...	3	PFC501283	DDO	0	0.0
2015/03/09 15:4...	2	PFC502487	DDO	0	0.0
2015/03/09 15:4...	1	PFC123456	DDO	0	55.5
2015/03/09 15:4...	4	PFC854321	DDO	0	0.0
2015/03/09 15:4...	3	PFC501283	DDO	0	0.0
2015/03/09 15:4...	2	PFC502487	DDO	0	0.0
2015/03/09 15:4...	1	PFC123456	DDO	0	55.5
2015/03/09 15:4...	4	PFC854321	DDO	0	0.0
2015/03/09 15:4...	3	PFC501283	DDO	0	0.0
2015/03/09 15:4...	2	PFC502487	DDO	0	0.0
2015/03/09 15:4...	1	PFC123456	DDO	0	55.5
2015/03/09 15:4...	4	PFC854321	DDO	0	0.0
2015/03/09 15:4...	3	PFC501283	DDO	0	0.0
2015/03/09 15:4...	2	PFC502487	DDO	0	0.0
2015/03/09 15:4...	1	PFC123456	DDO	0	55.5
2015/03/09 15:4...	4	PFC854321	DDO	0	0.0
2015/03/09 15:4...	3	PFC501283	DDO	0	0.0

Select Directory Export CSV Query